📘 PROJECT DOCUMENTATION

Full Stack Anime Management System with External API Integration using MERN Stack.

1. Abstract

The **Full Stack Anime Management System with External API Integration using MERN Stack** is a web-based application developed to manage users and fetch anime information dynamically from an external API.

The system integrates:

- React.js for frontend development
- Node.js & Express.js for backend services
- MongoDB for database management
- External Anime API (Jikan API)

The application allows users to register, login, manage users (CRUD operations), and search anime details in real-time. The project demonstrates full stack integration, RESTful API development, and external API consumption in a modern web application.

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2. Introduction

In today's digital world, full stack applications are widely used to integrate multiple technologies into a single system. This project demonstrates the implementation of a complete MERN stack application that performs:

- User Authentication
- User Management
- External API Integration
- Database Operations
- Responsive UI Design

The application ensures smooth interaction between frontend, backend, database, and third-party services.

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3. Objectives

The main objectives of this project are:

- To develop a full stack web application using MERN stack
- To implement user authentication (Register & Login)
- To perform CRUD operations using MongoDB
- To integrate an external API for anime data
- To create a responsive and user-friendly interface
- To demonstrate REST API architecture

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4. Technologies Used

4.1 Frontend

- React.js
- Axios
- Fetch API
- LocalStorage

4.2 Backend

- Node.js
- Express.js
- MongoDB
- Mongoose
- Axios

4.3 External API

- Jikan API (Anime Data Source)

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5. System Architecture

User
↓
React Frontend
↓
Express Backend (REST API)
↓
MongoDB Database
↓
External API (Jikan API)

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6. System Modules

6.1 Authentication Module

Features:

- User Registration
- User Login
- Token generation (Sample Token)
- LocalStorage token handling
- Logout functionality

Flow:

1. User enters credentials
2. Backend verifies user
3. Token returned
4. Token stored in LocalStorage
5. Protected pages displayed

6.2 User Management Module

This module allows admin/user to:

- Add User
- View Users
- Update User
- Delete User

CRUD Operations:

Operation	Method	Endpoint
Create	POST	/users
Read	GET	/users
Update	PUT	/users/:id
Delete	DELETE	/users/:id

6.3 Anime Search Module

This module integrates external anime data.

API Used:

https://api.jikan.moe/v4/anime?q=naruto

Features:

- Search anime by name
- Display title, image, synopsis
- Show score and episodes
- Show genres
- Display additional anime details
- Option to store anime data in MongoDB

Backend Logic:

- Receive query parameter
- Call Jikan API
- Map response
- Send structured JSON to frontend

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7. Database Design

7.1 User Schema

Fields:

- name (String)
- email (Unique)
- password
- role (default: user)
- timestamps

7.2 Anime Schema

Fields:

- title
- image (URL)
- synopsis
- score
- episodes
- genre (Array)

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8. Application Flow

1. User opens application
2. If token not found → Login page shown
3. After login → Dashboard loads
4. User can manage users
5. User can search anime
6. Anime results displayed dynamically
7. Logout removes token

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9. Features of the Project

- Full Stack Implementation
- RESTful API architecture
- MongoDB integration
- External API integration
- Dynamic data rendering
- Modern UI design
- Secure route handling
- Modular component structure

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10. Security Considerations

Current Implementation:

- Plain text password storage
- Sample token used

Recommended Improvements:

- Use bcrypt for password hashing
- Implement JWT authentication
- Add middleware protection
- Role-based authorization
- Input validation
- Environment variables

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11. Advantages

- Scalable architecture
- Clear separation of frontend and backend
- Easy maintenance
- Real-time external data fetching
- Responsive design
- Demonstrates real-world MERN stack integration

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12. Future Enhancements

- JWT based authentication
- Password encryption
- Role-based admin control
- Pagination for anime results
- Favorites system
- Deployment to cloud
- API caching
- Dark/Light theme toggle

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13. Conclusion

The **Full Stack Anime Management System with External API Integration using MERN Stack** successfully demonstrates the development of a complete web application integrating frontend, backend, database, and external services.

This project highlights:

- Practical implementation of MERN stack
- REST API development
- Database schema design
- Third-party API integration
- Full CRUD operations
- Authentication and session handling

The system provides a strong foundation for developing real-world full stack applications.